1 Basics of Networks and Electric Circuits

INTRODUCTION

In electronics engineering, generally, we want to transfer energy from one circuit to another. For this purpose, we require the interconnection of electrical and electronic components.

Such an arrangement of components is called an electric circuit, and each component of the interconnection is referred to as an element.

So, we can say that an electrical circuit is a combination of electrical/ electronics elements.

Basic quantity and their units:

S.No.	Quantity	Basic Unit	Symbol
1.	Length	Meter	m
2.	Mass	Kilogram	kg
3.	Time	Second	s
4.	Electric Current	Ampere	A
5.	Temperature	Kelvin	K
6.	Charge	Coulomb	C

Table 1.1

BASIC TERMINOLOGIES

1) **Circuit:** An interconnection of electrical elements such that a closed path is available for the flow of the current/charges.
For example:

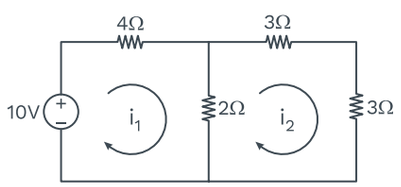


Fig. 1.1 Circuit

Note: Current can circulate in a closed path and hence, it is a circuit.

2) **Network:** Interconnection of network elements which may or may not have a closed path available is called network.

Definitions

An **electric circuit** is an interconnection of electrical devices, which will form a closed path through which the electric current can flow.

For example:

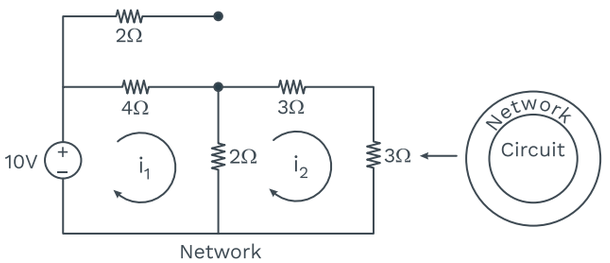
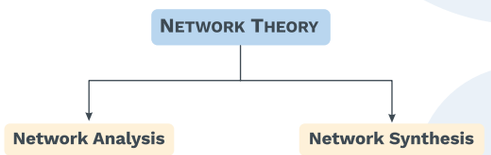


Fig. 1.2 Network circuit

Note: Circuit must have a closed loop path, but the network may or may not have a closed-loop path.

- Every circuit is a network, but every network is not a circuit.

Network theory is divided into two parts:



I) Network Analysis:



For a given input and network, the process of finding out the response (current or voltage or power) is known as network analysis.

II) Network Synthesis:

For a given input and response, the process of finding out the network is known as network synthesis.



Rack Your Brain

What do you mean by electric networks and circuits?

Grey Matter Alert!

The SI prefixes:

Multip-liers	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deca	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Table 1.2

CIRCUIT VARIABLES

Charge:

It is the fundamental property of any elementary particle by which particles interact with each other. It gives us knowledge about whether a particle has fewer or more electrons than protons. It describes how a particle acts in an electric field. It is measured in coulombs.

The smallest amount of charge that exists is the charge carried by an electron, equal to -1.6×10^{-19} coulomb.

The charge of a proton = $+1.6 \times 10^{-19}$ coulomb

- Charge is measured in coulomb.

  One coulomb = $\frac{1}{1.602 \times 10^{-19}}$ electrons

  = 6.24×10^{18} electrons
- According to the experimental observations, charges can exist in nature only as an integral multiple of the electronic charge, e = -1.602×10^{-19} C.
- The law of conservation of charge states that "total amount of charge produced in any process must always be zero" or "charge can be transferred from one object to another but it cannot be created or destroyed".

For example:

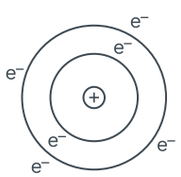


Fig. 1.3 Conservation of Charge

- Number of protons = Number of electrons
- Net charge = 0

  If an atom loses an electron, a positive charge dominates, i.e. total charge, $Q_{net} > 0$

  If an atom gains an electron, a negative charge dominates, i.e. total charge, $Q_{net} < 0$

Definitions

Charge: Electric charge is an inherent physical property of a particle. It is measured in Coulombs (C).

The charge of an electron is -1.6×10^{-19} C

The charge of a proton is $+1.6 \times 10^{-19}$ C

Rack Your Brain

The algebraic sum of the charges in a system remains constant. Give the reason?

Current:
Electric current is defined as the rate of flow of electric charge through any conductor having some cross-sectional area.
It is measured in amperes (A).
It is mathematically written as:

$$i(t) = \frac{dQ(t)}{dt} \text{ or } I = \frac{Q}{t}$$

Where
Q = charge (in coulomb)
t = time (in seconds)
Unit of current is coulomb/second or Ampere

Example 1:
How much charge is carried by 5000 electrons?

Sol:
Each electron has a charge of -1.6×10^{-19}C
Here given that number of electrons = 5000
Total charge by 5000 e^- = -1.602×10^{-19}C/e^- × 5000 e^-
= -8.010×10^{-16} C
The correct answer is -8.010×10^{-16} C

Example 2:
If the total charge of a terminal is $q = 10t \sin 4\pi t$ mC.
Find the current at t = 0.5 sec.

Sol:

We know $i(t) = \frac{dq}{dt}$

$$i(t) = \frac{d}{dt}(10t \sin 4\pi t) \text{ mC / sec}$$

$i(t) = (10 \sin 4\pi t + 40\pi t \cos 4\pi t)$ mA

$i|_{t = 0.5} = 10\sin 2\pi + 20\pi\cos 2\pi$

$i = 10 \times 0 + 20\pi \times 1$

$i = 20\pi = 20 \times 3.14$ mA

$i = 62.8$ mA

Hence, the value of current at t = 0.5 sec is 62.8 mA.

Definitions

The coulomb is the largest unit of charge. In one coulomb of charge, there are $1/(1.602 \times 10^{-19}) = 6.24 \times 10^{18}$ electrons.
Thus, the laboratory values of charges are in the order of pC, nC or μC.

Definitions

Electric Current: Electric current is the rate of the flow of charge in a conductor. Unit of current is coulomb/second or Ampere

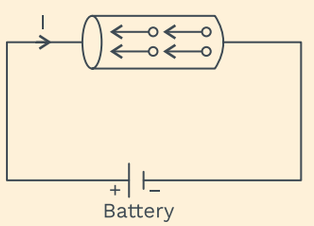


Fig. 1.4 Flow of charge in a conductor

Rack Your Brain

Is electric current a scalar or a vector quantity?

REFERENCE DIRECTION FOR CURRENT

The direction of current flow is the same as the movement of positive charge in any material. Figure 1.5 shows the convention that we use to describe a current. Here current i_1 is produced when the the electric charge moves from left to right, i.e. from A to B and current i_2 is produced when electric charge move from right to left, i.e. from B to A. Both have the same magnitude but opposite in direction, i.e. $i_1 = -i_2$



Fig. 1.5 Current Convention in Network

A current can be completely described by a value (which can be positive or negative) and a direction (indicated by an arrow).

Note:
For example, a current of 10A may be indicated with a positive or negative magnitude, as shown in figure 1.6 below.
Both the direction of current are indicated individually.

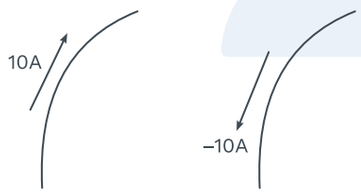
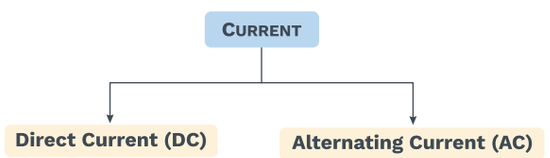


Fig. 1.6 Conventional Current Flow

Here, both the current have the same magnitude and are opposite in the direction.

Types of current:



Rack Your Brain

Why does current flow in the opposite direction to the flow of electrons?

Direct current:

- If a current has constant amplitude through all instants of time, then such current is known as direct current. It is depicted in figure Fig. 1.7 a) given below.



Fig. 1.7 a) Direct Current (DC)

Definitions

A direct current (DC) has constant amplitude over all the time.

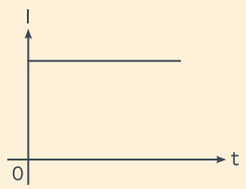


Fig. 1.8 Direct current

Alternating current:

- If a current has time-varying amplitude, then such current is known as alternating current. It is depicted in figure 1.7 b) given below.

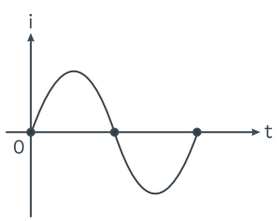


Fig. 1.7 b) Sinusoidal Current (AC)

VOLTAGE

The work done to move a unit positive charge from infinite to some reference point is known as electromotive force or potential or voltage.

$$V = \frac{dW}{dq} \text{ J / C or V}$$

$$V = \frac{W}{Q} \text{ J / C or V}$$

Where
W = Energy in Joules (J),
q = Charge in Coulombs (C)
and voltage V = measured volts (V).
1 Volts = 1 Joule/Coulomb
= 1 Newton-meter/Coulomb

Definitions

An alternating current (AC) has sinusoidally varying amplitude with time.

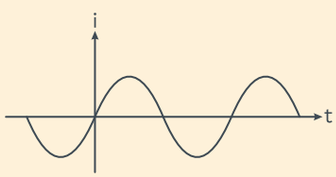


Fig. 1.9 Alternating Current

Thus, voltage is a representation of electric potential per unit charge.

- To move the electrons from one point to another point in a network requires an external force, which is provided by the electromotive force.
- Electric field intensity is:

$$E = -\nabla V \text{ or } E = -\frac{\Delta V}{\Delta l}$$

Reference polarity for voltage:

Figure 1.10 shows the voltage across an element (represented by a rectangular block) connected to points A and B. The plus (+) and minus (–) signs are used to define reference direction or voltage polarity. The V_{AB} can be interpreted in two ways:

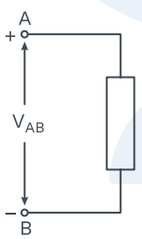


Fig. 1.10 Voltage Polarity Across an Element

1) Point A has a potential of V_A volts higher than point B. OR

2) Potential at point A with respect to point B is V_{AB}.

In general, it follows:

$$\left.\begin{aligned} V_{AB} &= -V_{BA} \\ V_{AB} &= V_A - V_B \\ V_{BA} &= V_B - V_A \end{aligned}\right\}$$

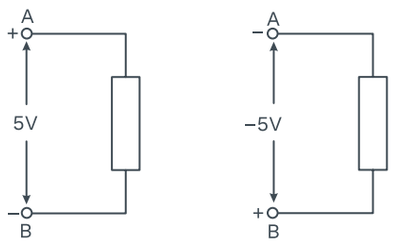


Fig. 1.12 Different Voltage Polarities Across an Element

Definitions

Voltage: "The amount of energy that an electric field can take to move unit of charge, measured in joule per coulomb".
1 volt = 1 Joule/Coulomb
= 1 Newton-meter/Coulomb

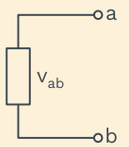


Fig. 1.11 Voltage source

Types of voltages:
Types of voltages are:

- **DC Voltage:** A voltage that remains constant with time is referred to as dc voltage.
- **AC Voltage:** A voltage that varies sinusoidally with time is referred to as ac voltage.

Note: Electric current always flows through an element, and the electric voltage is always across the element.

Keep in mind: Voltage does not exist at a point by itself; it is always determined with respect to some other point. For this reason, voltage is also called the potential difference. We often use the terms interchangeably.

Example 3:
A constant current of 4A is supplied by an energy source which will flow through a bulb for 10 seconds. Evaluate the voltage drop across the bulb when 4.6 kilo-Joule is given in the form of light and heat energy?

Sol:
Given, i = 4A
Δt = 10 seconds
The total charge = 4 × 10 = 40 C
The voltage drop is $V = \frac{\Delta W}{\Delta q}$

$$V = \frac{4.6kJ}{40}$$

$$V = \frac{4.6 \times 10^3}{40}$$

$$V = 115V$$

Rack Your Brain

What kills voltage or current?

ENERGY

Work done by an external agent in moving a charge from one point to another is provided as energy to charge.

- Capacity to do work is known as energy.

$$P = \frac{dW}{dt}$$

$$W = \int P \cdot dt$$

$$W = \int v \cdot i \, dt \quad \text{Joules}$$

Or

$$P = \frac{W}{t}$$

$$W = P \cdot t$$

$$W = v \cdot i \cdot t \text{ (“time in seconds”)}$$

$$W(t) = \int_{t_0}^{t} P(t)\, dt$$

Definitions

Energy: It is the ability to do the work. It is measured in Joules (J). The electric power utility companies measure energy in watt-hours (Wh), where

1 Wh = 3600 Joules

- Energy is expressed in watt-seconds or Joules (J).
- Energy can also be measured in watt-hours (Wh)

1Wh = 3600 Joules

Keep in mind: The electric bill that we pay to electric utility companies is paid for electric energy consumed over a certain period.

Example 4:
Find the energy consumed by a 100W electric bulb in 4 hours?

Sol:

W = Pt = 100(W) × 4 (h) × 60 (min/h) × 60 (sec/min)
W = 1440000 J = 1440 kJ
W = 1440 kJ

This is the same as,

W = Pt
W = 100 W × 4h
W = 400 Wh

Example 5:
A mobile phone has a battery of 12V and it is suitable for 10 minutes of talking time. While talking, it delivers 2A current and potential drop by 2V as shown below. Find the energy delivered by the battery during talking time.

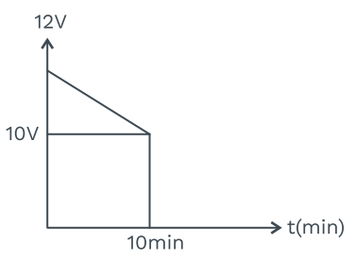


Fig. 1.13

Sol:

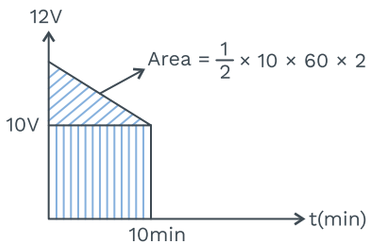


Fig. 1.14

$t = 10 \times 60 = 600$ seconds

$$\int V \cdot dt = 10 \times 10 \times 60 + \frac{1}{2} \times 10 \times 60 \times 2$$

$$\int V \cdot dt = 6600 \text{ volt-seconds}$$

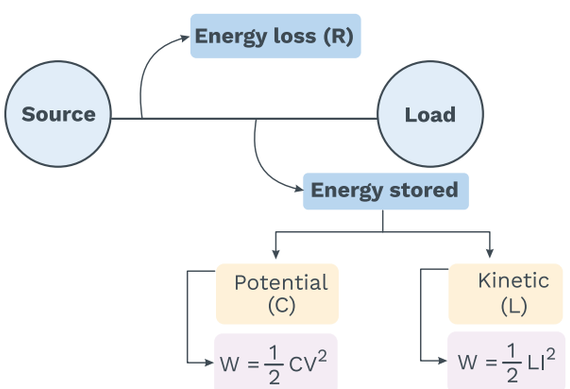
Energy $W = \int V \cdot i\,dt$

$$W = i \int V\,dt$$

$W = i \times 6600$
$W = 2 \times 6600$
$W = 13200$
$W = 13.2kJ$

Answer is 13.2 kJ

Keep in mind:



POWER:

The rate of change of energy is known as power.

- Power is the time rate of expanding or absorbing energy.
- It is measured in watts. Thus, in terms of energy, power is defined as:

$$P = \frac{dW}{dt}$$

$$P(t) = \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = v(t) \cdot i(t) \qquad \text{.... i)}$$

$$P(t) = v(t) \cdot i(t) \text{ Watts}$$

- We see that power is simply the product of voltage across an element and current through it.

Sign convention for power calculation:

- Current direction and voltage polarity are important while calculating the power convention.
- According to passive sign convention, the current enters the circuit element at the positive (+ve) terminal of the voltage and exits at the negative (–ve) terminal, as shown in figure 1.15 a).
- In this case, power absorbed by the circuit element is:

$$P = VI$$

- This power is also called power dissipated by the element or power received by the element.
- If the current enters the circuit element at the negative (–ve) terminal of the voltage and exit at the positive (+ve) terminal as shown in figure 1.15 b), the power absorbed will be:

$$P = -VI$$



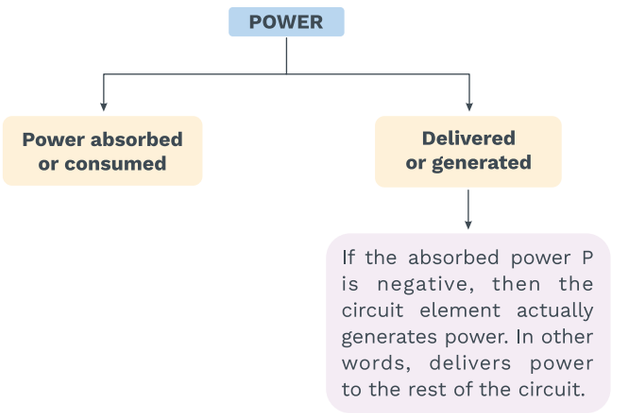
a) Absorbing power b) Supplying power

Fig. 1.15 Power Absorbing or Supplying by an Element

Definitions

Power is the rate of expanding or absorbing energy per unit time, measured in watt (W).

$$P = \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = VI \quad \text{Watts}$$



- In figure 1.7, however, the element is absorbing power of –20W but is equivalent to a supplying power of +20W.

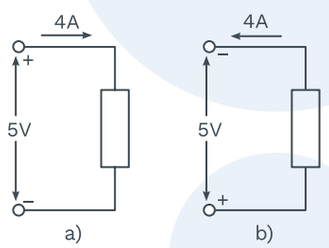


Fig. 1.16 Element with Absorbing Power

a) P = 5 × 4 = 20W
b) P = 5 × 4 = 20W

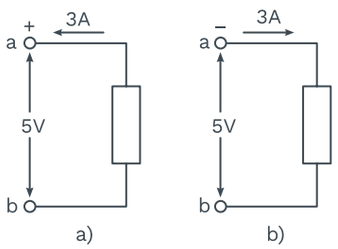


Fig. 1.17 Two Cases of an Element with a Supplying Power of 15W

a) P = –5 × 3 = –15W
b) P = –5 × 3 = –15W

- From the above explanation, it can be concluded that:

[Power delivered = – Power absorbed]

LAW OF CONSERVATION OF ENERGY:

The law of conservation of energy states that "The total energy of an isolated system is constant, and the total sum of power in the circuit, at any instant of time, must be zero.

$$\sum P = 0$$

- Thus, a sum of the absorbed power is always equal to the sum of delivered power in a circuit.

Mathematically we can write,

$$\sum P_{absorbed} = \sum P_{supplied}$$

Example 6:

If the current of a terminal is given by i = 5 cos 60πt, calculate the power delivered to element at 3 msec for i) v = 3i, ii) $v = 3\frac{di}{dt}$.

Sol:

i) The voltage is v = 3i = 15 cos 60πt

Hence, the power is:

$P = Vi = 75 \cos^2 60\pi t$ W

At t = 3msec, $P = 75 \cos^2 (60\pi \times 3 \times 10^{-3})$

$P = 75 \cos^2 0.18\pi = 53.48$

P = 53.48 W

ii) We find the voltage and the power as

$$v = 3\frac{di}{dt} = 3(-60\pi)5\sin 60\pi t$$

$v = -900\pi \sin 60\pi t$ volts

$P = Vi = -4500\pi \sin 60\pi t \cos 60\pi t$ W

At t = 3msec

$P = -4500\pi \sin 0.18\pi \cos 0.18\pi$ W

P = −6.396 kW

Previous Years' Question

A 12V automobile battery is connected to a 5A headlight system. The energy consumed in 1 hour of operation is _____kJ.

CIRCUIT ELEMENTS

The element is considered the basic building block for an electrical circuit. A simple circuit element is a mathematical model of a two-terminal electrical device, and it is characterized by its voltage-current relationship.

It cannot be subdivided into other two-terminal devices.

For example:

- If the voltage is directly proportional to the current in any element, then that element is said to be a resistor.



Fig. 1.18 Resistor

- If the terminal voltage is proportional to the derivative of current with respect to time, then the element is called an inductor.
- If the terminal voltage is proportional to the integral of current with respect to time, then the element is called a capacitor.
- If the terminal voltage is completely independent of current, or the current is completely independent of voltage, then the element is called as an independent source.
- The element for which either the voltage or current depends upon a current or voltage elsewhere in the circuit; such elements are called as dependent source.

Types of elements:

i) Active and passive elements
ii) Linear and non-linear elements
iii) Unilateral and bilateral elements
iv) Lumped and distributed elements
v) Time variant and time-invariant elements

Definitions

Active network: A network which contains circuit element with the energy source is called an active network.

i) Active and passive elements:

Active elements:

- The elements which are capable of generating a signal or amplifying a signal or rectifying a signal are known as active elements.

For example:
The voltage source, current source, operational amplifier etc.

Keep in mind: If we have a network element that is delivering power, i.e. energy delivered to the element $\left(\int_{-\infty}^{t} v(t)\,i(t)\,dt \right)$ is negative, then the element is an active element.

Passive elements:

- The elements that are not capable of generating a signal are known as passive elements.

For example:
Resistor, capacitor, diodes, inductor, etc.

Keep in mind: If we have a network element that is absorbing power, i.e. the energy delivered to the element $\left(\int_{-\infty}^{t} v(t)\,i(t)\,dt\right)$ is positive then the element is a passive element.
For example:
Resistor, inductor, diodes and capacitor.

Active Elements	Passive Elements
Voltage source	Resistor
Current source	Inductor
Transistor	Diodes
Operational-amplifier	Capacitor
SCR	Transformer

Table 1.3

Note: The active elements can provide power to the circuit or provide power gain in the circuit.

Remember:

- The transistors provide power gain, so they are active elements, but transformers have the same power at input and output. Thus, they are passive elements.
- The active element should be able to provide power, power gain to the circuit for an infinite duration of time; that is why the charged capacitor or inductor is not an active element.

ii) Linear and non-linear elements:

Linear elements:

- If an element satisfies the principle of homogeneity and additivity, then the element is said to be linear.
- **Linearity property:** This property is a combination of both the homogeneity (scaling) property and the additivity property.

Definitions

Passive network: A network which contains circuit element without any energy source is called a passive network.

- **Homogeneity property:** It states that if the input is multiplied by a constant, then the output must also get multiplied by the same constant.

  For example, if for excitation (voltage or current) $E_1(t)$, we get response (voltage or current) $R_1(t)$, then for excitation $E_2(t)$ we will get response $R_2(t)$.

Difference between linear and non-linear elements:

Linear Elements	Non-linear Elements
These elements are also called "Ohmic" elements.	These elements are also called "Non-Ohmic" elements.
These elements have a linear relationship between current and voltage.	These elements do not have linear input/output relations.
These elements obey the superposition theorem.	These elements do not obey the superposition theorem.
Example: Resistor, capacitor, inductor, transformer etc., are examples of linear elements.	Example: Most semiconductor devices have non-linear characteristics such as SCR, triac, etc.

Table 1.4

Additivity property:

- It states that the output to a sum of inputs is equal to the sum of the responses when individual inputs are applied.

  For example:

  If for excitation (voltage or current) $E_1(t)$ we get the response (voltage or current) $R_1(t)$ and for excitation (voltage or current) $E_2(t)$ we get response (voltage or current) $R_2(t)$, then for excitation $E_1(t) + E_2(t)$ we get response (voltage or current) $R_1(t) + R_2(t)$.

- An element that follows additivity and homogeneity properties for a relationship between excitation and response is called a linear element.

- An element that does not follow additivity and homogeneity property for a relationship between excitation and response is called a non-linear element.

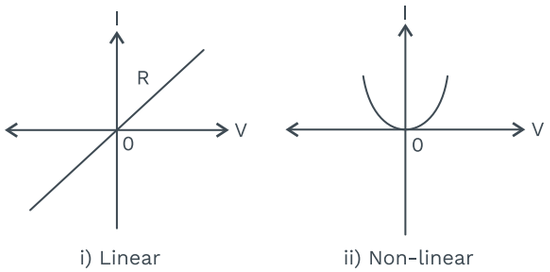


Fig. 1.19 Linear and Non-linear

Difference between linear and non-linear elements.

Linear	Non-linear
• Resistor	• Diode
• Inductor	• SCR, UJT
• Capacitor	• Transistor, op-amp
• Linear dependent sources	• Initially charged capacitor and inductor

Table 1.5

iii) Unilateral and bilateral elements:

Unilateral elements:

- For a unilateral element, the current is allowed to flow only in one direction. For example: Diode allows current flow from anode to cathode only.

Bilateral elements:

- For a bilateral element, the voltage-current a relationship is the same for current flowing in either direction.
- For example: resistors, inductors and capacitors allowing current in both the directions

Difference between unilateral elements and bilateral elements:

Unilateral Elements	Bilateral Elements
Conduction of current in one direction is termed unilateral.	Conduction of current in both directions with the same magnitude.
Examples of unilateral elements: Diode, Transistor etc.	Examples of bilateral elements: Resistors, Inductors, Capacitors

Table 1.6

iv) Lumped and distributed elements:

Lumped elements:

Lumped elements are considered as separate elements which are very small in size. For example: resistors, inductors and capacitors.

For example: The experiment in the laboratory.

Distributed Elements:

Distributed elements are not electrically separable. These are distributed over the entire length of the circuit. For example: transmission lines.

Keep in mind:

- Circuit theory is only applicable to the lumped system, but electromagnetic field theory is applicable for both lumped and distributed systems.
- The size of the lumped element is small with respect to signal wavelength. At steady-state, we can consider distributed elements as lumped element.

Previous Years' Question

Which of the following code satisfy a non-linear network?

1) Associative
2) Superposition
3) Homogeneity
4) Bilaterality

Select the correct answer using the codes given below:

a) 1 and 3 only
b) 1 and 4 only
c) 2 and 3 only
d) 2 and 4 only

v) Time variant and time invariant elements:

- **Time variant:** The element whose characteristics are dependent upon a time is known as time variant element.
- **Time invariant:** The element whose characteristics are not dependent upon time is known as time invariant element.

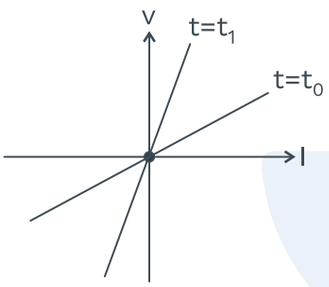


Fig. 1.20 Characteristics Change due to time

Definitions

Bilateral element: Elements which allow the flow of current through them in both directions.

I-V CHARACTERISTIC CURVES FOR DIFFERENT ELEMENTS:

Following are given some I-V characteristic curves for different elements, looking at these characteristics, we can find the type of element.

Characteristics:

1)

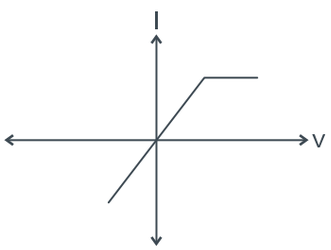


Fig. 1.21 I-V Characteristics

i) Non-linear
ii) Uni-directional
iii) Passive
{1st& 3rd quadrant are not the same}

- If the current direction and voltage polarity are not specified, then by means of a slope, it can be identified as active.

Rack Your Brain

Give some practical examples where lumped and distributed circuit elements are observed?

2)

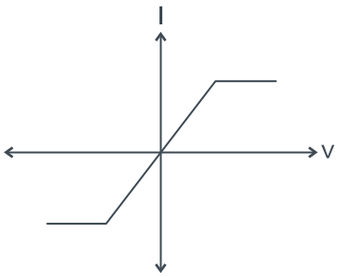


Fig. 1.22 Characteristics

i) Non-linear **ii)** Bidirectional **iii)** Passive

3)

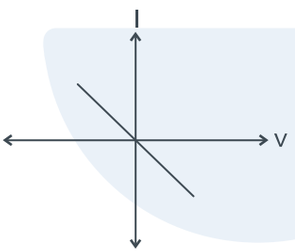


Fig. 1.23 Characteristics

i) Linear **ii)** Bidirectional **iii)** Active

4)

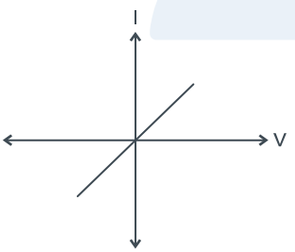


Fig. 1.24 Characterstics

i) Linear **ii)** Bidirectional **iii)** Passive

5)

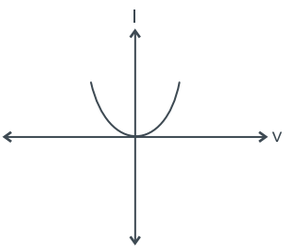


Fig. 1.25 Characteristics

i) Non-linear **ii)** Unidirectional **iii)** Active

6)

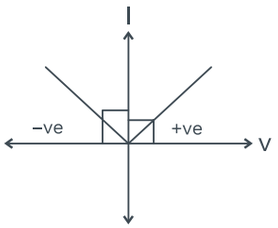


Fig. 1.26 Characteristics

i) Non-linear
ii) Unidirectional
iii) Active

Keep in mind:

- Passive elements do not have negative (–ve) slope characteristics.
- $W(t) = \int_{-\infty}^{\infty} v(t)\,i(t)\,dt$ < 0 Active, > 0 Passive
- If $\frac{V}{I}$ or $\frac{I}{V}$ is positive (+ve) $\rightarrow$ Passive
- If negative (–ve) $\rightarrow$ Active

  e.g.

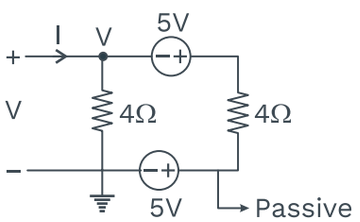


Fig. 1.27 Circuit

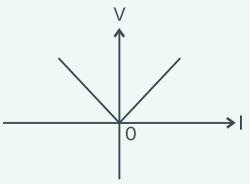
$$I = \frac{V}{4} + \frac{V + 5 - 5}{4}$$

$$I = \frac{V}{2}$$

i) Passive
ii) Linear
iii) Bilateral

Previous Years' Question

The v-i characteristics of an element is shown in the figure given below. The element is:



a) Non-linear, active, non-bilateral
b) Linear, active, non-bilateral
c) Non-linear, passive, non-bilateral
d) Non-linear, active, bilateral

Rack Your Brain

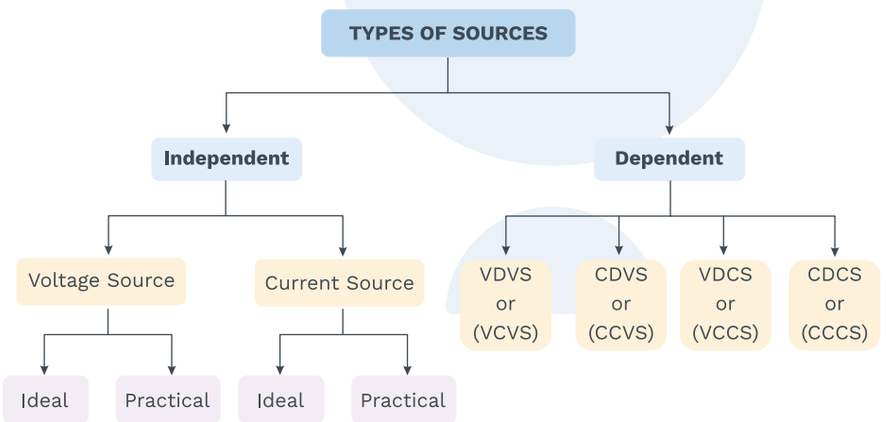
What are the applications of constant voltage sources?

OBSERVATIONS

- All linear elements are bilateral but bilateral elements need not be linear.
- If the characteristic curve is similar in opposite quadrants, then the element is bidirectional; otherwise, it is unidirectional.
- If the ratio of voltage to current at any point on the characteristics curve is negative, then the element is active; otherwise, it is passive.
- Every linear element must exhibit bidirectional property.

SOURCES

In this section, we introduce a basic circuit element called a source. Sources are classified as voltage sources and current sources.



Where
VDVS-Voltage dependent voltage source
VCVS-Voltage controlled voltage source
CDVS-Current dependent voltage source
CCVS-Current controlled voltage source
VDCS-Voltage dependent current source
VCCS-Voltage controlled current source
CDCS-Current dependent current source
CCCS-Current controlled current source

Independent voltage source:
An independent voltage source is characterized by a terminal voltage which is completely independent of the current through it.
The circuit symbol is shown in figure 1.28; the subscript (s) merely identifies the voltage as a “source” voltage and is common but not required.

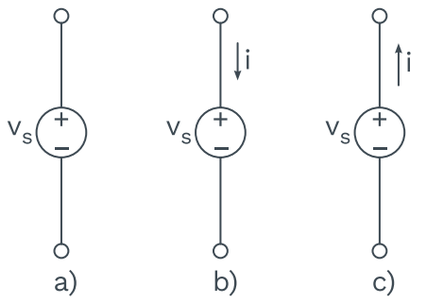


Fig. 1.28 Circuit Symbol of the Independent Voltage Source

Rack Your Brain

Why transformer is not an active element?

Ideal voltage source:

- Ideal voltage source is defined as a voltage source whose voltage at any instant of time is independent of current through its terminals.

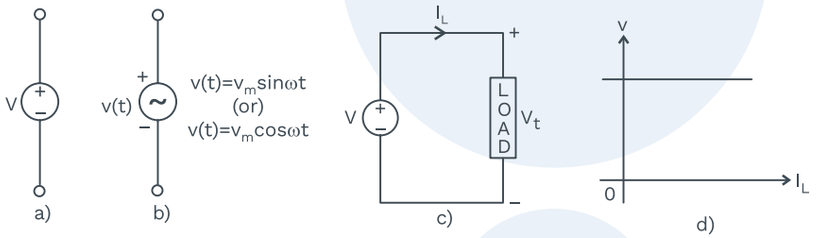


Fig. 1.29 Ideal Voltage Sources

- In the above graph terminal voltage line is parallel to the load current line so that the graph is non-linear & Ohm's law is not applicable across a voltage source.
- It is a unilateral element.
- In an ideal voltage source, voltage regulation is zero.

Grey Matter Alert!

Ideal independent sources can be batteries, DC generators or time-varying AC voltage supplies from alternators.

Practical voltage source:

- Practical voltage source is defined as a voltage source whose voltage is dependent upon current through its terminals.

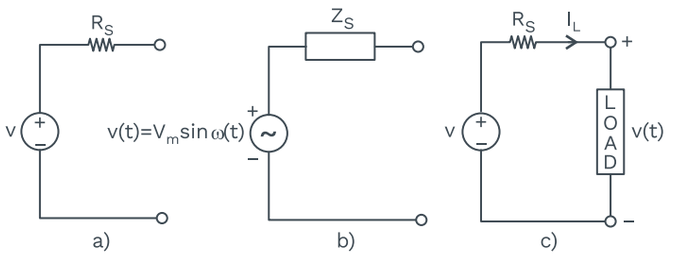


Fig. 1.30 Practical Voltage Sources

Here,

$v(t) = V_m \sin\omega t$

or $v(t) = V_m \cos\omega t$

R_s = Source or internal resistance

Z_s = Source or internal impedance

$V = I_L R_S + V_t$

$\downarrow V_t = V - I_L \uparrow R_S$

V_t = Terminal voltage

V = Source voltage

I_L = Load current

$$R_S = \frac{\text{Open circuit voltage}}{\text{Short circuit current}}$$

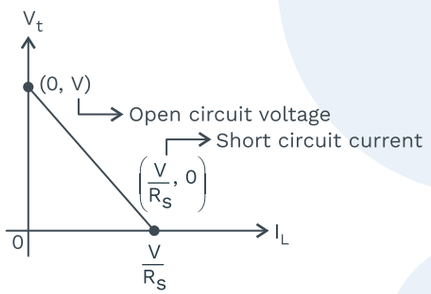


Fig. 1.31 Calculation of Open Circuit Voltage and Short Circuit Current

Independent current source:

Ideal current source:

An ideal current source is defined as a current source whose current at any instant of time does not depend upon the voltage across its terminal.

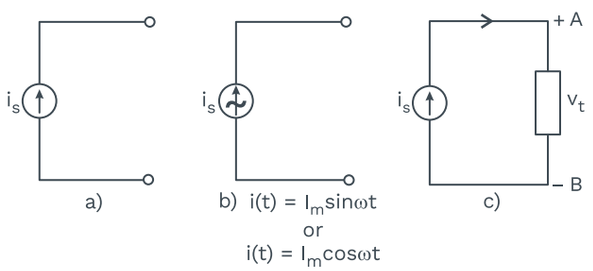


Fig. 1.33 Ideal Current Sources

- In an ideal current source, the current regulation is zero.

Previous Years' Question

Consider the following circuit:

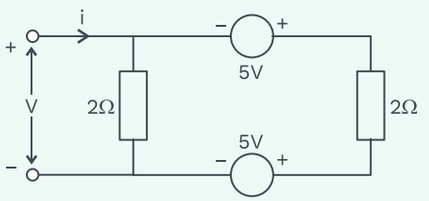


Fig. 1.32

Which one of the following statements is correct? The circuit shown above is:

a) Passive and linear
b) Active and linear
c) Passive and non-linear
d) Active and non-linear

Grey Matter Alert!

An ideal voltage source has zero internal resistance and an ideal current source has infinite internal resistance.

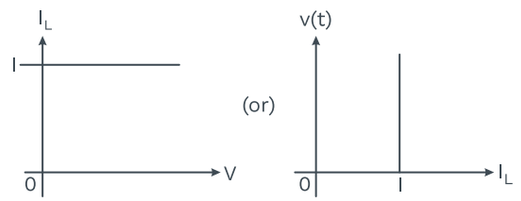


Fig. 1.34 Ideal Current Source or Voltage Source Characteristics

Practical current source:

- Practical current source is defined as a current source whose current at any instant of time depends upon the voltage across its terminals.
- In an ideal current source, internal resistance is $\infty\,\Omega$ or '0' mho.

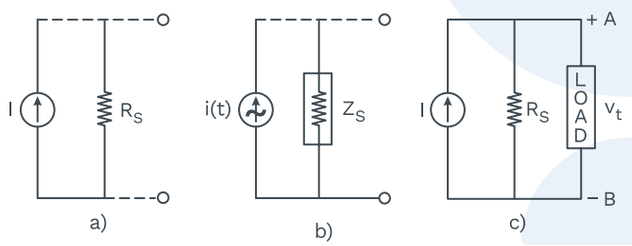


Fig. 1.35 Practical Current Sources

$$I = I_L + \frac{V_t}{R_S}$$

$$I_L \downarrow = I - \frac{V_t \uparrow}{R_S}$$

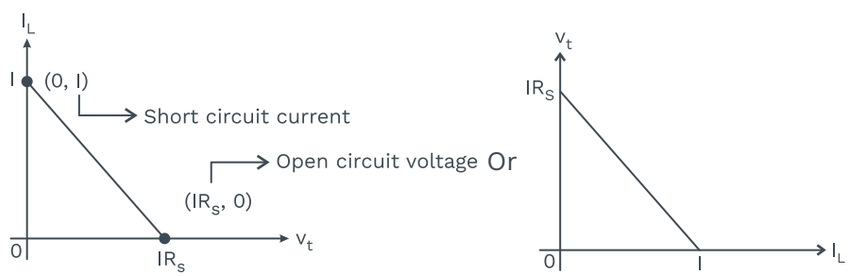


Fig. 1.36 Characteristics

$$R_S = \frac{\text{Open circuit voltage}}{\text{Short circuit current}}$$

Rack Your Brain

What is the difference between dependent and independent sources?

Previous Years' Questions

If an ideal voltage source and an ideal current source is connected in series, then the combination:

a) has the same properties as a current source alone

b) has the same properties as a voltage source alone

c) has the same properties as the source which has a higher value

d) results in the branch being redundant

Keep in mind: An independent sources supply power to the circuit. However, they can be connected into a circuit in such a way that they absorb power from the circuit.

Note: Symbolic representation of voltage source and current source.

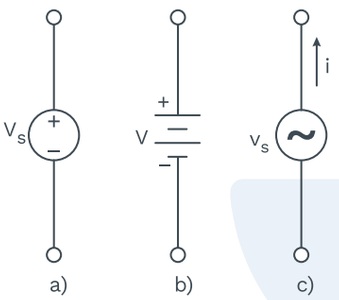


Fig. 1.37 a) DC Voltage Source Symbol, b) Battery Symbol, c) AC Voltage Source Symbol

- An independent current source with a constant terminal current is often termed as independent dc current source and can be represented by the symbol shown in figure a).
- The symbol for an independent ac current source is shown in figure b).

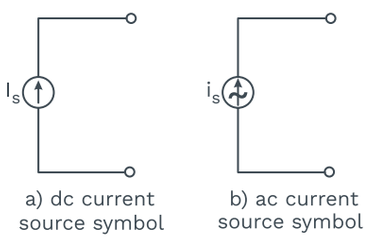


Fig. 1.38

Keep in mind:

- It is a common mistake to view an independent current source as having zero voltage across its terminals while providing a fixed current. In fact, we do not know what the voltage across a current source will be; it depends entirely on the circuit to which it is connected.

Rack Your Brain

Compare ideal voltage source with the practical voltage source.

Rack Your Brain

Compare current source with voltage source.

Grey Matter Alert!

In an ideal voltage source, voltage regulation is zero and in an ideal current source, current regulation is zero.

- The presence of the plus sign at the upper end of the symbol of the independent voltage source in the figure does not necessarily mean that the upper terminal is numerically positive with respect to the lower terminal. Instead, it means that the upper terminal of V_s volts is positive with respect to the lower.
- If at some instant, V_s is negative, then the upper terminal is actually negative with respect to the lower at that instant.

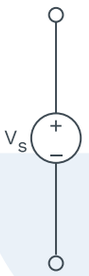


Fig. 1.39 Symbol of a Voltage Source

Dependent sources:

Dependent sources are the sources whose output depends upon some other voltage or current in a circuit. Both voltage and current types of sources may be dependent and may be controlled by a voltage or a current. Thus, there are four types of dependent sources:

1) Voltage-controlled voltage source (VCVS); $v = Av_x$
2) Current-controlled voltage source (CCVS); $v = Ai_x$
3) Voltage-controlled current source (VCCS); $i = Av_x$
4) Current-controlled current source (CCCS); $i = Ai_x$

Rack Your Brain

Does a constant current source exist? If yes, what are the practical examples?

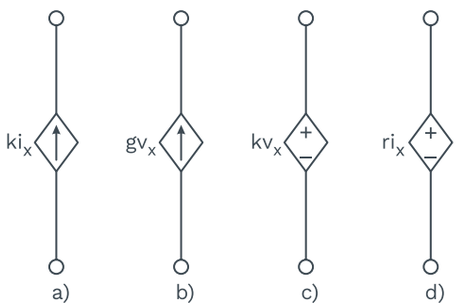


Fig. 1.40 The Four Different Types of Dependent Sources

a) **Current-controlled current source**
b) **Voltage-controlled current source**
c) **Voltage-controlled voltage source**
d) **Current-controlled voltage source**

Keep in mind:

- A different symbol, in the shape of a diamond, is used to represent dependent sources.
- Dependent sources are very useful in describing certain types of electronic circuits.
- A dependent source may absorb or supply power.

Note: A dependent source works as an active source if the circuit contains at least one independent source and the dependent variable is not equal to zero.



Rack Your Brain

Do dependent current and voltage sources exist practically?

Chapter Summary



- **Circuit:** An interconnection of electrical elements such that a closed path is available for the flow of the current or charges.
- **Charge:** Charge is an electrical property of the atomic particles, measured in Coulombs (C).
- **Current:** Electric current is the time rate of change of charge, measured in amperes (A).

$$i(t) = \frac{dq(t)}{dt} \text{ Amperes}$$

- **Voltage:** Voltage (or potential difference) is the energy required to move a unit charge through an element measured in volts (V).

$$V = \frac{dW}{dq} \text{ volts}$$

- **Power:** Power is the time rate of expanding or absorbing energy, measured in watts (W).

$$P(t) = \frac{dW}{dt} \text{ Watts}$$

- **Energy:** Energy is the capability to perform work. The energy over a time interval is found by integrating the power.

$$W(t) = \int P(\tau)\, d\tau$$

- **Active and passive elements:** The elements that are capable of generating a signal are known as active elements, otherwise passive elements.

  Active: Voltage source, current source, transistor.

  Passive: Resistor, Inductor, capacitor and transformer.
- **Linear and non-linear elements:** The elements that obey superposition and homogeneity properties are known as linear elements, otherwise non-linear elements.
- **Bilateral and unilateral elements:** The elements that offer the same impedance in both directions of current are known as bilateral elements, otherwise unilateral elements.

  Bilateral elements: Resistors, inductors and capacitors.

  Unilateral elements: A diode is a unilateral element.

- **Lumped and distributed elements:** If the elements are physically separable, then they are called lumped elements, otherwise distributed elements.

  Lumped elements: Experiment in a laboratory.

  Distributed elements: Long transmission line, Antenna.
- **Time variant and time invariant:** Elements whose characteristics are dependent upon a time are known as time variants, otherwise time invariant.

IMPORTANT CHARACTERISTICS:

1)

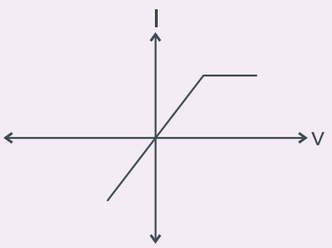


Fig. 1.41 Characteristics

i) Non-linear **ii)** Unidirectional **iii)** Passive

2)

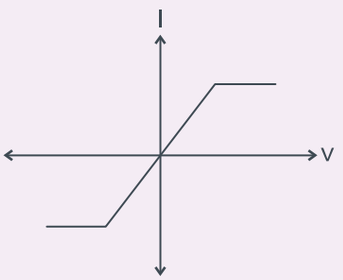


Fig. 1.42 Characteristics

i) Non-linear **ii)** Bidirectional **iii)** Passive

3)

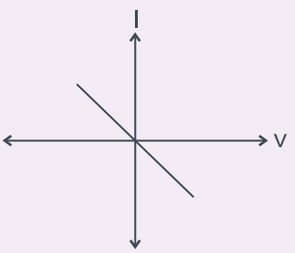


Fig. 1.43 Characteristics

i) Linear **ii)** Bidirectional **iii)** Active

4)

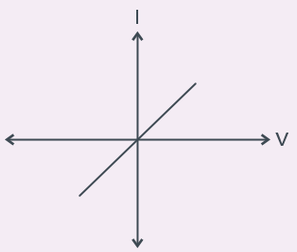


Fig. 1.44 Characteristics

i) Linear **ii)** Bidirectional **iii)** Passive

5)

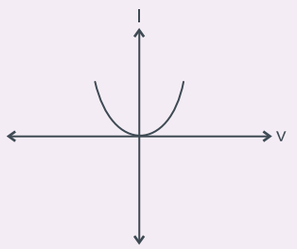


Fig. 1.45 Characteristics

i) Non-linear **ii)** Unidirectional **iii)** Active

6)

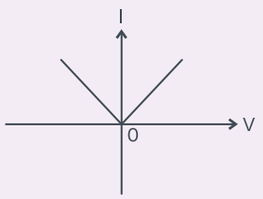


Fig. 1.46 Characteristics

i) Non-linear **ii)** Unidirectional **iii)** Active

Ideal Sources:

1) Ideal voltage source:
The voltage always remains constant for any value of current passing through it.

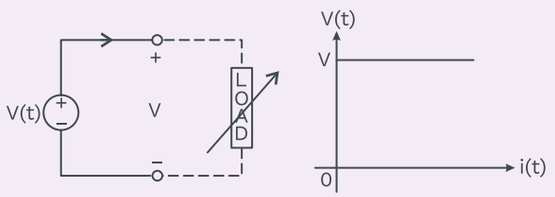


Fig. 1.47 Ideal Voltage Source

2) Ideal current source:
The current always remains constant for any value of voltage across it.

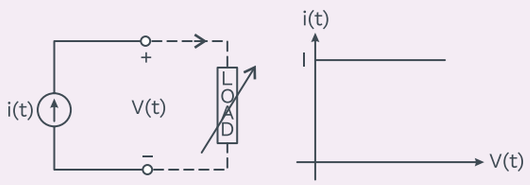


Fig. 1.48 Ideal Current Source

- Dependent Sources:

a)  $v(t) = Ki_1 \Rightarrow$ current controlled voltage source (CCVS)

b)  $v(t) = Kv_2 \Rightarrow$ voltage controlled voltage source (VCVC)

c)  $i(t) = Kv_1 \Rightarrow$ voltage controlled current source (VCCS)

d)  $i(t) = Ki_2 \Rightarrow$ current controlled current source (CCCS)

where K is a constant.

PRACTICE QUESTIONS

Q1 **Which of the following amount of electrons is equivalent to –3.941 C of charge?**
a) 1.628×10^{20} b) 1.24×10^{18} c) 6.482×10^{17} d) 2.46×10^{19}

Q2 **A solid copper sphere, 10 cm in diameter, is deprived of 10^{20} electrons by a charging scheme. The charge on the sphere is _____ C.**

Q3 **A charge of 120 C passes through a conductor in 60 seconds. The current in the conductor is _____ (in Ampere).**

Q4 **The following plot shows the charge entering a certain element. The value of current at t = 0.5 sec, 3 sec and 5 sec are respectively.**

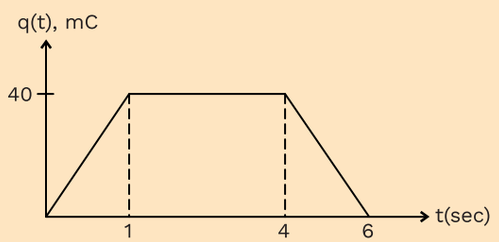


Fig. 1.49

a) 40 mA, 0 mA, –20 mA **b) 40 mA, 40 mA, 20 mA**
c) 40 mA, 0 mA, 20 mA **d) –40 mA, 0 mA, 20 mA**

Q5 **The charge passing through a circuit element is sketched in the figure below.**

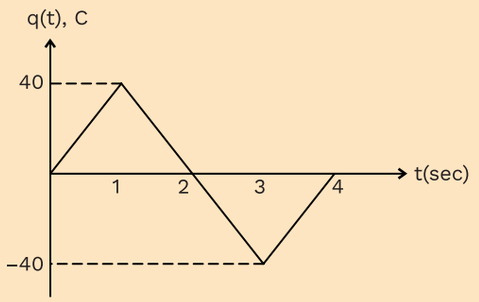
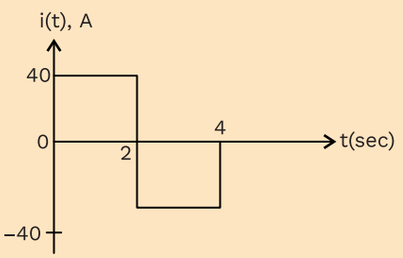


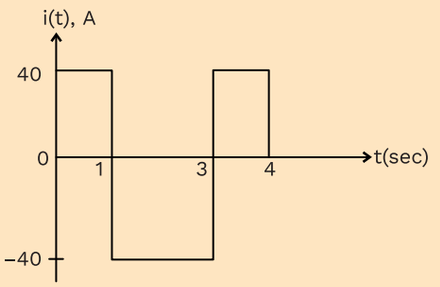
Fig. 1.50

The plot of the current i(t), flowing through the element, will be:

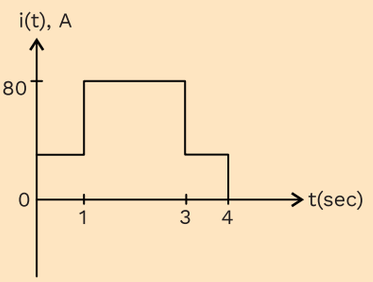
a)



b)



c)



d)

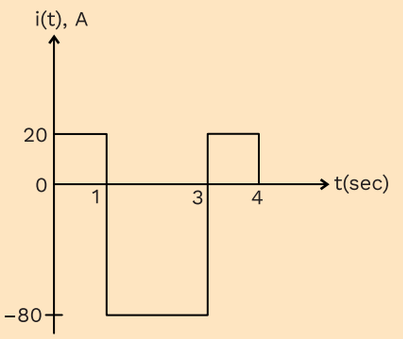


Fig. 1.51

Q6 **A circuit consisting of four circuit elements is shown in the figure below. The voltage drop and polarity are indicated in the figure. It is given that V_{AB} = 8V, then calculate the value of V_1?**

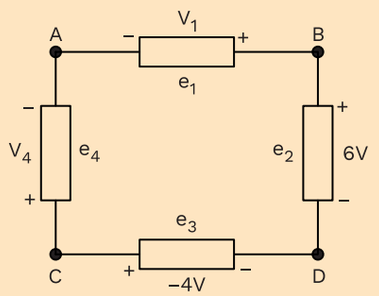


Fig. 1.52

a) 8V **b) −8V** **c) −8V** **d) 8V**

Q7 **In figure shown below, when 12 coulombs of charge pass through the element from A to B, the energy absorbed by the element is 60J. What is the voltage V across the element?**

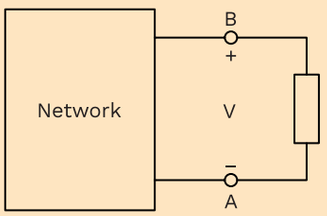


Fig. 1.53

Q8 **The energy required to move 120 coulombs through 3V is _____Joule.**

Q9 **In the circuit shown below, power P is:**

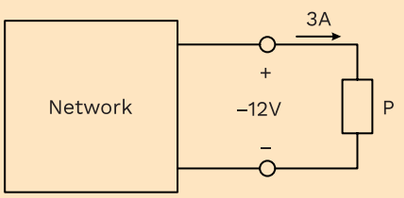


Fig. 1.54

a) 36W, absorbed
b) 18W, supplied
c) 36W, supplied
d) None of these

Q10 **If the current and voltage for a terminal is given as i = 5 cos 60πt and $v = 3\frac{di}{dt}$. Calculate the power delivered to an element at t = 3 ms.**

a) –6.396 kW
b) 5 kW
c) 57.48 kW
d) 3 kW

Answer key:

1)	**d)**	**2)**	**16.02**	**3)**	**2 Ampere**	**4)**	**a)**	**5)**	**b)**
6)	**b)**	**7)**	**–5 volt**	**8)**	**360 Joule**	**9)**	**c)**	**10)**	**a)**